

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

- Ph.D. Seoul National University**, Mathematical Sciences Mar. 2021 – Feb. 2026
- Advisor: [Ernest K. Ryu](#)
- B.S. Korea University**, Mathematics Education Mar. 2015 – Feb. 2021
- Graduated 1st in Class
 - (Mandatory) Military Service: May 2017 - Jan. 2019
 - Secondary School Teacher Certificate (Grade II), Mathematics

Experience

- Staff Engineer**, Samsung Electronics Suwon, South Korea Mar. 2026 -
- Mobile eXperience
 - Developing on-device Q&A models for Bixby on the Galaxy S series.
- Research Internship**, CryptoLab Seoul, South Korea Dec. 2025 - Feb. 2026
- Homomorphic Encryption Optimization Team
- Workshop Coordinator**, SNU-KAIST ML Theory Workshop Gangneung, South Korea Aug. 2024
- Assisted in planning the schedule, selecting the venue, and managing workshop programs.
- Teaching Practicum** Seoul, South Korea Apr. 2020 - May 2020
- Dangdok high school
- Military Service**, Auxiliary Police Seoul, South Korea May 2017 – Jan. 2019
- Assisted in maintaining public order and security.

Publications and Preprints

*: Equal Contribution

- [4] **On the Low Rank Theory of Data Shapley: A Kernel Perspective** Sep. 2026
Seungchul Shin*, **Donghwan Rho***, Myungjoo Kang
NeurIPS 2026 ATTRIB Workshop
- [3] **Bridging LLMs and Symbolic Phonological Rules: RLVR for the Korean Word-Chain Game** Oct. 2025
Donghwan Rho
In progress
[Paper](#)

- [2] **Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models** Oct. 2025
Donghwan Rho*, Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
 Under review
[Paper](#) [Code](#)
- [1] **Encryption-Friendly LLM Architecture** Oct. 2024
Donghwan Rho*, Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae,
 Ernest K. Ryu, Jung Hee Cheon
 International Conference on Learning Representations (ICLR) 2025
[Paper](#) [Code](#)

Talks & Presentations

- Symposium of the Research Institute of Mathematics, Seoul National University,** Nov. 2025
 Talk
 • Title: Implementing Efficient Language Models under Homomorphic Encryption
[Site link](#), [Blog post](#) (Thanks to [Yossi Frenkel](#)), [Slides](#)
- Korean Mathematical Society Annual Meeting,** Talk Oct. 2025
 • Title: Homomorphically encrypted language models stabilized with TSP-sorted tokens
[Link](#)
- SNU-Samsung Electronics Collaboration Symposium,** Poster Presentation Aug. 2024
 • Title: Encryption-Friendly LLM Architecture
- SNU-KAIST ML Theory Workshop,** Poster Presentation Aug. 2024
 • Title: Encryption-Friendly LLM Architecture

Academic Service

- Conference Reviewer**
- NeurIPS (2026)
 - ICLR (2026)
 - NeurIPS Workshop (2026)

Honors and Awards

- Outstanding TA,** Seoul National University [Link](#) July 2025
 Introduction to Computer Programming and Artificial Intelligence for Scientists
- University Students Contest of Mathematics,** Silver Prize [Link](#) Dec. 2018
- Academic Excellence Scholarship,** Korea University Aug. 2015

Teaching

- Peer Tutoring in Basic Sciences,** Lead Teaching Assistant Mar. 2024 - Feb. 2026
 • Represented section TAs to ensure the smooth operation of the peer tutoring program.
 • Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.
- Basic Calculus,** Teaching Assistant Sep. 2021 - Feb. 2026
 • Matched tutors and tutees for the *Basic Calculus* peer tutoring program.
 • Managed program logistics to ensure efficient operation.

SNU Pre-College Program, Head TA, Instructor

- Managed logistics of the mathematics program for freshman undergraduate.
- Gave lectures.
- Winter 2025, Winter 2024, Winter 2023, Winter 2022

Introduction to Computer Programming and Artificial Intelligence for Scientists,
Instructor

Spring 2025

- English Course
- Selected as an [outstanding TA](#) 

Topics in Machine Intelligence

Fall 2024

- Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models
- English Course
- Graduate Course

Mathematical algorithms II

Fall 2022

- Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing
- English Course
- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub